

Rohan Sada

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SUMMARY

AI/ML Engineer with 3+ years of production experience architecting scalable deep learning systems, computer vision pipelines, and distributed training infrastructure across GPU clusters. MS in Electrical & Computer Engineering from Rutgers University specializing in machine learning and sensor fusion. First-author IEEE-published researcher with expertise in LLM fine-tuning, RAG systems, and large-scale distributed training; recipient of the AllGoVision Excellence Award.

EXPERIENCE

AllGoVision Technologies

Apr 2022 – Aug 2024

AI Vision Engineer

Bengaluru, India

- Architected a multi-threaded C++/Python inference architecture that scaled deep-learning-driven multi-camera tracking to **100+ concurrent** high-resolution RTSP streams in production.
- Overhauled cross-camera re-identification by engineering spatio-temporal matching (Extended Kalman Filter + Dynamic Time Warping) on top of learned embeddings, **doubling (2x)** re-identification accuracy while pruning low-probability candidates to cut compute.
- Optimized deep learning inference pipelines with NVIDIA TensorRT kernel fusion, cutting per-frame latency **30%** and unlocking real-time analytics at scale; containerized and shipped the pipeline for remote client-site production deployment.

WINLAB, Rutgers University

May 2025 – Aug 2025

Research Intern

New Brunswick, NJ

- Architected an end-to-end streaming data pipeline integrating a spatial-temporal LSTM network to forecast urban mobility demand, optimizing inference to serve real-time predictions across a live interactive map engine at **90% accuracy**.
- Engineered an automated active-learning ingestion pipeline that processed **1,000+ hours** of raw video; implemented a pseudo-labeling framework that eliminated **85%** of manual annotation bottlenecks and accelerated downstream model retraining cycles.

PROJECTS

DataLens – Compound AI Text-to-SQL Platform | [GitHub](#)

FastAPI, Qdrant, Neo4j, Redis, Qwen2.5-Coder

- Architected a compound AI Text-to-SQL pipeline (FastAPI backend) supporting DB upload, SQL-script import, and live credential-based connections across SQLite/Postgres/MySQL, returning natural-language answers with executable SQL and tabular results.
- Engineered a dual vector + graph schema-retrieval router (Qdrant cosine-similarity search + Neo4j foreign-key graph expansion) that adaptively serves full vs. retrieved schema context, cutting prompt-token consumption on large databases without degrading SQL-generation accuracy.
- Fine-tuned Qwen2.5-Coder-7B via QLoRA (SFT) and GRPO (Group Relative Policy Optimization) on the BIRD Text-to-SQL benchmark, then closed the reliability loop with self-correcting SQL retries (up to 3 on DB error) against a read-only, AST-validated sandboxed execution engine.
- Deployed the system as a horizontally-scalable, Redis-backed, multi-service architecture (FastAPI + Next.js + Streamlit) instrumented with Prometheus/structlog observability and validated by a pytest CI suite.

3D Object Detection Extension for SLAM3R | [GitHub](#)

PyTorch, ViT, DETR, Hungarian Matching

- Engineered a DETR-style 3D detection head attached to the frozen SLAM3R Vision Transformer backbone, training only **8.4%** of total parameters to jointly produce 3D reconstruction and object detections in a single forward pass.
- Curated a 13,000+ image multi-view indoor dataset across ten categories and trained the head with Hungarian matching and a custom 3D GloU loss, achieving **51% mAP@25** on 3D object detection.

Distributed BLIP-2 Fine-Tuning | [GitHub](#)

PyTorch FSDP, DeepSpeed ZeRO-3, SLURM

- Fine-tuned Qwen 2.5 VL on the ROCO medical image-captioning dataset using PyTorch FSDP (full parameter/gradient/optimizer sharding, bf16 mixed precision) and DeepSpeed ZeRO-3 across **16 A100 GPUs** on the NERSC Perlmutter supercomputer.
- Diagnosed CUDA out-of-memory failures crashing under-sharded 2-GPU configurations, then re-engineered full-shard parameter distribution to cut peak memory **34%** and unlock stable 16-GPU training, measuring a **10.7x throughput speedup**.
- Benchmarked DDP, FSDP, and DeepSpeed ZeRO-3 across 4/8/16-GPU multi-node topologies, pushing ZeRO-3 to **91% average GPU utilization** and demonstrating super-linear scaling for sharded strategies versus sub-linear DDP scaling.

EDUCATION

Rutgers University

Sep 2024 – May 2026

MS, Electrical & Computer Engineering, GPA: 3.8/4.0

New Brunswick, NJ

PES University

Graduated 2022

B.Tech, Electrical & Computer Engineering, CGPA: 8.11/10.0

Bengaluru, India

TECHNICAL SKILLS

Machine Learning & Deep Learning:

LLM Fine-Tuning (QLoRA, GRPO), RAG Systems, Computer Vision, Model Optimization, MLOps

Distributed Training Infrastructure:

PyTorch FSDP, DeepSpeed ZeRO-3, DDP, Multi-Node SLURM Orchestration, CUDA

Frameworks:

PyTorch, TensorFlow, HuggingFace Transformers, Scikit-learn, LangGraph, TensorRT, ONNX

Data & Retrieval Systems:

Qdrant (Vector DB), Neo4j (Graph DB), Redis, PostgreSQL, MySQL, SQLite

Languages:

Python, C++, C, JavaScript, SQL

PUBLICATIONS

R. Sada, D. T. M., and R. Rajesh, "Deep Learning Based Real-Time Fusion of Day and Thermal Videos," *IEEE TENSYP 2022*, Mumbai, India, pp. 1–6. DOI: [10.1109/TENSYP54529.2022.9864438](https://doi.org/10.1109/TENSYP54529.2022.9864438)